

The Radiomics–Microbiome Axis Layer: A Hybrid Text–Vision Pipeline with Five-Gate Acceptance for Knowledge-Graph Extension

Junseong Lee

SanoMap Project — Radiomics-Microbiome Axis Layer

junseong.lee652@gmail.com

Draft of May 19, 2026 (pre-print; benchmark metrics gated, see §4.5)

Abstract

Background. Literature-mined microbiome knowledge graphs such as the SanoMap MINERVA platform connect microbes directly to diseases but omit the quantitative, tissue-level imaging phenotypes that mechanistically link them. Much of the quantitative microbe–imaging correlation evidence is additionally locked inside figures, invisible to text-only pipelines. **Methods.** We build a radiomics-first intermediary layer that inserts `RadiomicFeature` and `BodyCompositionFeature` nodes between microbe and disease. A two-track pipeline (a text track over 1,016 open-access papers; a vision track that reads correlation heatmaps with a vision-language model) feeds a five-gate acceptance model: dense-retrieval candidate generation, UMLS type-grounded entity sanitization, self-consistency relation acceptance, dual-verifier vision consensus behind a deterministic pre-verifier gate chain, and a stratified gold-label benchmark. Accepted edges are reconciled into a single provenance-stamped graph export and loaded into Neo4j with a read-only query layer. **Results.** The graph carries 74 phenotype–disease `ASSOCIATED_WITH` edges, 29 signed microbe–disease edges, and—after a retroactive vision audit and a graph reconciliation that composed both the vision retraction and the UMLS drop onto one base—7 `CORRELATES_WITH` microbe–feature edges (1 vision, 6 text), closing 62 traversable microbe→feature→disease paths a single-hop schema cannot express. The vision track’s measured contribution is deliberately scoped to one audited case-study figure plus a hallucination-resistant gating methodology. Precision/recall and intra-annotator Cohen’s κ are gated on a non-negotiable 14-day temporal re-labeling window and are reported as pending, not estimated. **Conclusion.** Making the imaging phenotype explicit is feasible, reviewable, and reproducible; the engineering contribution is the acceptance-gate architecture and an honest accounting of where each evidence modality does and does not deliver.

Keywords: knowledge graph; radiomics; gut microbiome; vision-language models; biomedical relation extraction; Neo4j; evidence provenance.

1. Introduction

Biomedical knowledge graphs derive their clinical value from the completeness of the evidence they encode. Transformer-based language models and generative AI have enabled large-scale construction of such graphs by systematically mining verified literature. The SanoMap project’s MINERVA platform (Microbiomes Network for Research and Visualization Atlas) demonstrates this well: it maps associations between gut microbes and systemic diseases at scale, producing a clinically navigable graph from thousands of curated papers.

Yet MINERVA’s graph encodes a structurally incomplete picture. It connects microbes directly to diseases but omits the quantitative, tissue-level intermediate phenotypes that explain the biological mechanism linking them. Radiomics and body-composition imaging provide exactly these intermediaries. Skeletal muscle index, visceral adipose tissue fraction, liver fat fraction, and GLCM texture signatures are extracted directly from CT and MRI and are established biomarkers for sarcopenia, nonalcoholic fatty liver dis-

ease, and colorectal cancer. The literature increasingly demonstrates that gut microbiome composition measurably influences these specific tissue-level phenotypes—making them natural intermediary nodes in any complete microbiome–disease graph.

A second structural gap concerns the evidence modality itself. Text-mining pipelines extract findings authors highlight in prose, but much of the quantitative correlation data linking microbial taxa to imaging feature values is encoded in figures. Gradient correlation heatmaps display matrices of r -values linking dozens of taxa to radiomic measurements—data NLP alone cannot parse. This is a systematic evidence loss in text-only pipelines.

This work addresses both gaps. The central research questions are: **(RQ1)** Can a hybrid pipeline combining large language models and vision-language models, with deterministic verification, reliably extract quantitative correlation coefficients directly from medical figures? And **(RQ2)** does adding radiomics as an ontological layer—filling the microbe→imaging-phenotype→disease path—meaningfully increase the clinical completeness

of the MINERVA knowledge graph? A third, engineering question runs underneath both: **(RQ3)** what acceptance architecture keeps such a hybrid graph *reviewable*—auditable edge-by-edge—rather than merely large?

This paper’s contribution is not a claim of upstream parity. It is (i) an explicit intermediary-layer ontology, (ii) a five-gate acceptance model that converts a proof-of-concept pipeline into a measurable artifact, (iii) a retroactive audit that honestly rescopes the vision track after a structural verifier weakness was found, and (iv) a single reconciled, provenance-stamped graph reproducibly loadable into Neo4j.

2. The Intermediary Layer: Concept and Schema

2.1 Core idea

MINERVA’s current graph encodes (Microbe)→[ASSOCIATED_WITH]→(Disease). This edge is clinically underspecified. The biologically accurate path is

Microbe → *correlates with* → Radiomic/BodyCompFeature
→ *associates with* → Disease.

This three-node path is documented concretely: *Prevotella nigrescens* correlates with GLCM Correlation texture on CT at $r = 0.95$ (PMC10605408); cirrhosis dysbiosis enriches Proteobacteria alongside measurable changes in liver surface nodularity and visceral fat; *Bifidobacterium* depletion co-occurs with elevated visceral adipose tissue and downstream obesity. These are quantitatively measured, directly evidenced relations that a text-only microbe→disease edge collapses into a single mechanism-blind link. Making the intermediary explicit enables queries impossible in MINERVA’s schema, e.g. “what CT texture signature mediates the association between gut dysbiosis and sarcopenia?”

2.2 Node and edge ontology

Node types extend MINERVA: **RadiomicFeature** (texture, morphological, intensity features—GLCM, wavelet, first-order, shape); **BodyCompositionFeature** (sarcopenia, visceral adipose tissue, skeletal muscle index, liver fat fraction, myosteatosis); **BodyLocation**; **ImagingModality**; **ImageRef** (verified figure references with Vision Track provenance); **Microbe / MicrobialSignature**; and **Disease**.

The direct-evidence edge policy is deliberately evidence-typed: quantitatively verified microbe→feature edges (a vision-confirmed r -value) use **CORRELATES_WITH**; text co-mention feature→disease edges use **ASSOCIATED_WITH**; signed microbe→disease edges use **POSITIVELY_/NEGATIVELY_CORRELATED_WITH**; structural edges use **MEASURED_AT**, **ACQUIRED_VIA**, **REPRESENTED_BY**. The

CORRELATES_WITH/ASSOCIATED_WITH distinction preserves evidence-level semantics across the graph and is never collapsed. Within-paper bridge matches that only share disease context are written to an audit artifact as hypotheses and are *never* ingested as asserted edges.

3. Methods

3.1 Data sourcing

Data was sourced from PubMed Central Open Access via a multi-lane query strategy over six clinical domains where the microbiome–radiomics link is best evidenced: (1) strict radiomics + microbiome; (2) body composition + microbiome; (3) liver radiomics + microbiome; (4) bone-density (DXA) + microbiome; (5) lung-CT phenotypes + microbiome; (6) colorectal imaging + microbiome. The corpus spans **1,016 papers** (640 from the initial three lanes plus 376 net-new from lanes 4–6), with open-access full text retrieved where a PMCID exists. Review, meta-analysis, and protocol papers are excluded as non-direct-evidence targets.

3.2 Two-track pipeline

The **Text Track** extracts phenotype–disease associations at scale. A vocabulary-matched stage maps imaging-phenotype terminology to canonical names; microbial NER uses **d4data/biomedical-ner-all** on Apple Silicon MPS, disease NER uses the **en_ner_bc5cdr_md** spaCy pipeline; a shared span-cleanup layer rejects wordpiece artifacts, generic placeholders, and clause-prefix fragments before relation classification.

The **Vision Track** addresses the figure-data gap. A vision-language model proposes r -values for microbe→feature correlations read from heatmap figures; each proposal is deterministically cross-checked against the figure’s gradient color-scale legend at pixel level.

3.3 Five-gate acceptance model

Tracks alone do not establish research-defensibility; structurally distinct gates do. A candidate edge enters the graph only when every gate that applies to its evidence track passes.

- 1. Retrieval (text).** A dense-retrieval index over **BioClinical-ModernBERT-base** embeddings (mean-pooled, L2-normalized; FAISS where available, NumPy cosine fallback at $\sim 30k$ sentences) replaces the original hand-curated 25-alias substring filter, removing its recall ceiling on paraphrastic mentions (“loss of muscle mass at L3”). Per-feature similarity threshold τ is calibrated per concept against labeled data.
- 2. Entity sanitization (UMLS TUI grounding).** Microbial surface forms must ground to a microbe-class UMLS Semantic Type at similarity ≥ 0.85 ; accept set {T005 Virus, T007 Bacterium, T194

Archaeon, T204 Eukaryote} with an explicit non-microbe model-organism deny-list. A live audit of the 8 accepted text-derived microbe→feature edges drops 25% (2/8): the entity-type error **gut bacterial ClpB-like gene function** (similarity $0.799 < 0.85$, grounded to “Bacteria” T007) and the typo **bacteriodetes** (no UMLS match). The 25% drop sits below the 30% recalibration threshold; the 0.85 floor is retained.

3. **Relation acceptance (text).** Gemini 2.5 Flash-Lite, 7-sample temperature-varied self-consistency, full agreement required; measured full-consistency rate on the expanded corpus is 0.916.
4. **Verification (vision) — dual verifier behind a pre-verifier gate chain.** See §4.3; this gate’s design is itself a contribution.
5. **Evaluation (gold benchmark).** A 5-stratum sampler (accepted_edge, gemini_rejected, vocab_excluded, recall_probe, random_co_occurrence) drawn deterministically from the production candidate space yields a 66-row gold set. The schema defines a 6-class primary label, three secondary labels, eight numbered edge-case decisions, and a 14-day intra-annotator temporal-relabeling protocol with Cohen’s $\kappa \geq 0.80$ (binary collapse) acceptance. Pass-1 used computer-aided manual annotation (LLM proposer, single human reviewer who overrode 7 rows under an imaging-derived scope rule); Pass-2 is the annotator’s independent re-label after the temporal window. Metrics that depend on Pass-1+Pass-2 are pending (§4.5).

3.4 Graph integration and reproducibility

Earlier iterations emitted several divergent CSV vintages and a hand-runnable Cypher script, with no single coherent “current graph.” This release adds a deterministic reconciliation step: one command reads the authoritative assembled artifacts, applies the documented post-audit drop-list (the retracted Firmicutes vision edge; the two UMLS-failed microbe→feature edges), and emits a single **graph_export/ bundle—nodes.csv, relationships.csv**, an idempotent **import.cypher**, and a **manifest.json** recording source-artifact vintages, the git commit, pre/post counts, and every drop with its reason. This step immediately earned its keep: it found that the pre-existing prose headline (“8 CORRELATES_WITH: 1 vision + 7 text”) had applied only the vision retraction to the 9-edge superset and never composed in the separately recorded UMLS drop; composing both onto one base yields the coherent 7 (1 vision + 6 text). The bundle loads into Neo4j through a thin driver loader; a read-only query module exposes the canonical traversals (three-hop microbe→feature→disease; feature-by-disease; signed microbe→disease; feature-by-location; full modality-backbone chain; vision-verified edges). Provenance, not just the graph, is the deliverable: every edge

resolves back to a PMID, sentence, or figure panel, and every count in this paper is reproducible from the manifest.

4. Results

4.1 Extracted graph

Across the 1,016-paper corpus the pipeline produced:

- **5,721** phenotype text mentions across 30 clinically meaningful disease targets (initial 640-paper corpus; expanded-corpus re-extraction in progress);
- **74** ASSOCIATED_WITH phenotype→disease edges (Gemini full-consistency 0.916 on the expanded corpus);
- **29** signed microbe→disease edges (14 positive, 15 negative) spanning cirrhosis, NAFLD, colorectal cancer, IBD, celiac disease, chronic HCV, and SSc-associated pulmonary fibrosis;
- **7** CORRELATES_WITH microbe→feature edges *after* reconciliation (§4.3): 1 vision (*Prevotella nigrescens* → GLCM_Correlation, $r = 0.95$, PMC10605408_g004; pixel distance 0.05, support fraction 1.0) and 6 text (Gemini 7/7);
- **18** BodyLocation and **5** ImagingModality nodes;
- **189** reconciled relationship rows over **99** nodes. The pre-reconciliation export reported 191 rows / 9 CORRELATES_WITH; the reconciliation step (§3.4) drops exactly two edges—one vision retraction, one UMLS-failed entity—each recorded in the manifest with its cited source artifact, rather than silently changing the headline.

Three surviving CORRELATES_WITH edges close end-to-end three-hop paths through features that also carry downstream ASSOCIATED_WITH disease edges—*Ruminococcus*→sarcopenia (37 diseases), *Peptostreptococcus stomatis*→skeletal_muscle_index (7), *Eubacterium*→visceral_adipose_tissue (18)—yielding **62 traversable microbe→feature→disease paths**, the query pattern MINERVA’s single-hop schema cannot express. This is direct evidence for RQ2: the intermediary layer adds traversable mechanism, not just nodes.

4.2 Corpus-level findings

Two findings merit note. *Faecalibacterium* depletion emerged as a convergent signal across six distinct disease associations in the new liver and colorectal lanes (chronic liver disease, colorectal cancer, IBD, celiac disease, IBS, SSc-associated pulmonary fibrosis), consistent with its established role as a butyrate-producing protective commensal. *Fusobacterium nucleatum* enrichment in colorectal cancer was independently confirmed across four PMIDs from the colorectal lane, validating that lane’s signal quality.

4.3 Vision-track audit and honest scoping

A retroactive audit of all vision inputs (13 current proposals + 1 historical graph edge) revealed that dual-verifier consensus *alone* is structurally insufficient: the

pixel HSV verifier and the independent VLM verifier both consume the proposer’s bounding box and proposed r -value, so a self-consistent fabrication (proposer hallucinates both fields so the predicted color matches the colorbar at that location on the assumed default ± 1.0 scale) passes AND-consensus silently. The pixel verifier also defaulted its observed range to $[-1, +1]$ rather than reading tick labels, so log-fold-change (± 1.5) heatmaps were validated as if Pearson r .

Three deterministic pre-verifier gates were added in `src/vision_gates.py` and run *before* consensus: a **caption gate** (requires Pearson/Spearman vocabulary; auto-fails log fold change, LFC, z-score, differential abundance); a **colorbar-detection gate** (a gradient legend must be present); and a **range-sanity gate** ($|r_{\text{prop}}| \leq \max(|c_{\text{min}}|, |c_{\text{max}}|) + 0.05$, with $(c_{\text{min}}, c_{\text{max}})$ read from the figure’s tick labels by a focused VLM call). On the audit batch ($n=14$) the chain rejected 6 figures (3 caption, 2 range-sanity, 1 colorbar); the dual verifier flagged 3 of the remaining 8 for human review. The historical *Prevotella nigrescens* edge was retained (caption-confirmed Spearman heatmap, extracted -1 to $+1$ colorbar, pixel distance 0.05, direct image inspection). The historical *Firmicutes*→Total fat % edge ($r = -0.95$, PMC6178902) was **retracted**: the asterisked cell is in the figure’s positive (red) hemisphere, the colorbar runs -1.5 to $+1.5$ (a log-fold-change scale, not Pearson r), and the prior “pixel-verified” pass was a circular consistency check on a hallucinated bounding box.

This reframes the vision contribution honestly. Rather than a pre-audit “9 edges (2 vision + 7 text)” framing, the reconciled graph carries 7 (1 vision + 6 text), once the vision retraction and the UMLS drop are composed onto one base (§3.4). The publishable vision result is therefore *not* edge volume; it is (a) one fully audited case-study figure and (b) a hallucination-resistant gating methodology with a documented residual failure mode—wrong-sign hallucination when the bounding box happens to point at a same-color cell elsewhere—whose cheapest mitigation (a sign-check gate comparing claimed sign vs. VLM-observed color hemisphere) is specified for future work. This is the honest answer to RQ1: deterministic verification is necessary but not sufficient; the gate chain, not the consensus, is what makes vision evidence defensible.

4.4 Verification and software quality

The pipeline carries 280 automated test checks across the full suite (up from 156 before the structural-upgrade work; +105 new tests cover the UMLS validator, dense retrieval, dual verifier, gold-set sampler, and benchmark evaluator), with the graph-export reconciliation and query layer added in this release additionally covered. No regressions on prior tests.

4.5 Gated metrics

Measured precision/recall and intra-annotator Cohen’s κ are deliberately reported as **pending**, not estimated. The benchmark requires a non-negotiable 14-day intra-annotator temporal window between Pass-1 (closed 2026-05-07) and Pass-2 (earliest 2026-05-21); a shorter interval leaks short-term memory and inflates κ artificially. Reporting an estimate here would violate the evidence-typed discipline the rest of the system enforces. On the window’s close we will report binary precision/recall/F1 with 95% Wilson confidence intervals (wider, $\sim \pm 0.10$, because the corpus-limited gold set is 66 rows, not the 150-row design) and Cohen’s κ on the two human passes.

5. Discussion: Comparison with MINERVA

The system extends MINERVA, not replicates it. Each gap below is an explicitly acknowledged design decision, not an error.

Gap 1 — NER models. MINERVA fine-tuned a custom DistilBERT (BNER2.0) on a proprietary corpus (F1=0.914; weights unreleased). We use `d4data/biomedical-ner-all` (microbe) and `en_ner_bc5cdr_md` (disease), selected via a controlled 5-paper benchmark. Residual risk: lower precision on phylum/genus/species disambiguation, partially compensated by span cleanup and a genus allowlist. Closure paths: Professor-mediated BNER2.0 access, or the public `OpenMed-NER-SpeciesDetect` (DeBERTa-v3+LoRA, SotA on Linnaeus/S-800) as a no-fine-tune upgrade.

Gap 2 — Relation model. MINERVA fine-tuned BioMistral-AUG-7B (weights unreleased). We use Gemini 2.5 Flash-Lite zero-shot with 7-sample self-consistency (full-consistency rate 0.896–0.916, a calibrated reliability measure, not a P/R claim). Closure path: fine-tune BiomedBERT on MicrobioRel (22 gut-microbiome relation types) or the Lever et al. 2023 labeled set.

Gap 3 — UMLS normalization: closed and exceeded. We match MINERVA’s CUI-linked deduplication (ScispaCy `en_core_sci_lg`, 2022 UMLS KB; 96% / 83% CUI coverage on the initial / expanded corpora) and go beyond it by promoting the same CUI/TUI meta-data to an *active acceptance gate*: a microbe failing to ground to a microbe-class TUI at similarity ≥ 0.85 is rejected pre-classification with an audited `drop_reason`.

6. Limitations and Future Work

Radiomic parameter abstraction. The pipeline captures modality and body location but abstracts away bin width, voxel resampling, slice spacing, and ROI method—rarely reported in mined text. Two GLCM entropy values from different bin widths become one node. Future work: map to IBSI feature-instance codes encoded

Table 1: Gap status relative to MINERVA.

Gap	Sev.	Status
NER (microbe)	Med	Substitute identified; TUI gate mitigates type errors
NER (disease)	Low	BC5CDR competitive; SotA upgrade identified
Relation model	Med	Self-consistency 0.916; fine-tune path documented
UMLS norm.	High	Closed & exceeded (TUI-gated, audit-active)
Text recall ceiling	Med	Closed (ModernBERT dense retrieval)
Vision monoculture	High	Partially closed: dual gate alone insufficient; 3 pre-verifier gates added; audit + reconciliation 9→7 edges
Measured P/R	High	Pending Pass-2 (§4.5)

ing extraction context.

Demographic heterogeneity. Body-composition metrics depend strongly on age, sex, and BMI; the graph aggregates across heterogeneous cohorts without a `PopulationContext` property. Future work: a cohort node or edge stratum enabling demographic filtering.

Inter-scanner variability — a vision-track advantage. By extracting r -values from heatmaps rather than absolute radiomic values, the vision track inherently normalizes across scanner hardware: a rank-order correlation is far more stable across sites than an absolute HU value, giving vision edges higher cross-site generalizability than value-anchored text associations.

Observational only. The graph encodes association, not causality. Future work: **Intervention** nodes (FMT, antibiotics, probiotics, diet) with typed `MODIFIES/REVERSES` edges, shifting toward a partially causal model—requiring longitudinal pre/post datasets targeted in future harvest-lane design.

Sparsity and track complementarity. The three evidence layers are complementary by construction. The text track (phenotype→disease) is a broad associative network (74 edges, 30 diseases); the text track (microbe→feature) contributes 6 self-consistency-validated `CORRELATES_WITH` edges; the vision track contributes 1 quantitatively verified edge plus the gating methodology. The microbe→feature edges are what close the 62 three-hop paths the intermediary-layer thesis requires; remaining `CORRELATES_WITH` edges without downstream disease edges will close as text-track phenotype extraction expands to additional feature vocabularies. The graph is intentionally sparse and reviewable rather than dense and unaudited—this is a design stance, answering RQ3.

7. Conclusion

Inserting an explicit radiomics–body-composition intermediary between microbe and disease is feasible, mechanistically meaningful, and—given the five-gate acceptance model and a single provenance-stamped graph export—reproducible and reviewable. The vision track’s honest contribution is a hallucination-resistant verification methodology and one audited case study, not edge volume; the benchmark’s headline metrics are gated on a disciplined temporal window and reported as pending rather than estimated. The system’s defensibility comes from where it refuses to overclaim as much as from what it extracts.

Reproducibility. Code, tests, the reconciled `graph_export/` bundle with its provenance manifest, the Neo4j loader, and the read-only query layer are tracked in the project repository. The pre-reframing single-column proposal report is preserved unmodified at `docs/proposal/archive/`. Every numeric claim here is reproducible from the export manifest; gated metrics will be appended on the temporal window’s close.